



CHAPTER

8

The p-Block Elements

Sr. No.	Concept	Description	Other information
1.	General electronic configuration of p-Block elements	ns^2np^{1-6}	
2.	Group 13 elements: (a) General electronic configuration (b) Metallic radii (c) General oxidation state	ns^2np^1 Do not increase regularly Common oxidation state = + 3	Other O.S. = +1
3.	Physical properties of group 13th elements (Boron family) (a) Atomic radii (b) Ionic radii (c) Electronegativity (d) Ionization enthalpy (e) Melting point (f) Boiling point (g) Density	$B < Ga < Al < In < Tl$ $Tl > In > Al > Ga > B$ $B > Tl > In > Ga > Al$ $B > Tl > Ga > Al > In$ $B > Al > Tl > In > Ga$ $B > Al > Ga > In > Tl$ $Tl > In > Ga > Al > B$	Size $\propto$ atomic radii M.P. decreases from B to Ga and then increase B..P. decreases down the group Density increases down the group
4.	Group 14 elements: (a) General electronic configuration (b) Oxidation state (c) Stability of O.S.	ns^2np^2 Common oxidation state = +4 (i) Stability of +4 O.S. $C > Si > Ge > Sn > Pb$ (ii) Stability of +2 O.S. $Pb > Sn > Ge > Si > C$	Other O.S. = +2, -2

Sr. No.	Concept	Description	Other information
5.	Physical properties of group 14th elements (Carbon family) (a) Covalent radii (b) Ionization enthalpy (c) Electronegativity (d) Melting point (e) Boiling point (f) Density	$C < Si < Ge < Sn < Pb$ $C > Si > Ge > Pb > Sn$ $C > Pb > Si \approx Ge \approx Sn$ $C > Si > Ge > Sn > Pb$ $Si > Ge > Sn > Pb$ $Si < C < Ge < Sn < Pb$	
6.	Some important compounds of Boron (a) Borax (b) Orthoboric acid (c) Diborane	$Na_2B_4O_7 \cdot 10H_2O$ H_3BO_3 B_2H_6	
7.	Some important compounds of Carbon & Silicones (a) Carbon monoxide (b) Carbon dioxide (c) Silicone dioxide (d) Silicones (e) Silicates (f) Zeolites	CO CO_2 SiO_2 $(R_2SiO)_x$ (SiO_4^{4-}) $Na_2Al_2Si_2O_8 \cdot xH_2O$	